

Schrödinger’s Navigator: Imagining an Ensemble of Futures for Zero-Shot Object Navigation

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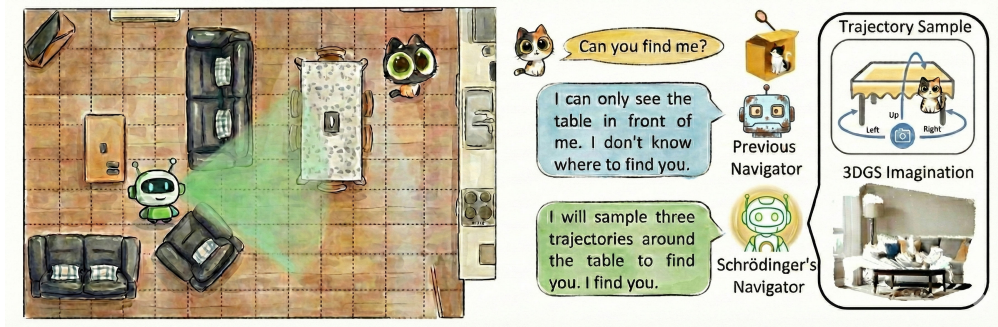


Figure 1: Real-world zero-shot object navigation often fails when the target (e.g., a cat) is hidden behind occlusions and surrounded by hazardous space. Conventional navigation systems perceive only the immediate occluder and are unable to infer what exists beyond it. Our Schrödinger’s Navigator addresses this challenge by modeling the unobserved regions as plausible futures.

Abstract: Zero-shot object navigation (ZSON) requires robots to find target objects in unseen environments without task-specific fine-tuning or pre-built maps, a key capability for general-purpose service robots. Yet methods that perform well in simulation often degrade in cluttered real-world scenes with severe occlusion and latent hazards, where large unseen regions make single-scene inference brittle and unsafe. We propose **Schrödinger’s Navigator**, a belief-aware framework that reasons at inference time over multiple *trajectory-conditioned imagined 3D futures*. Given candidate paths, a trajectory-conditioned 3D world model predicts hypothetical observations and maintains a superposition of plausible scene realizations rather than committing to one map. An adaptive occluder-aware sampler directs imagination to uncertainty-critical regions, while a *Future-Aware Value Map* (FAVM) aggregates imagined futures for robust, proactive action selection. Experiments in simulation and on a physical Go2 quadruped show that Schrödinger’s Navigator outperforms strong ZSON baselines, improving hidden-target discovery and risk-aware waypoint selection in occlusion-heavy navigation scenarios. These results highlight imagined 3D futures as a scalable and generalizable strategy for zero-shot navigation in uncertain real-world environments.

Keywords: Zero-Shot Object Navigation, Generative Scene Imagination

1 Introduction

Object navigation is a fundamental capability for mobile robots operating in human-centered environments [1, 2, 3]. Practical service robots must find requested objects in previously unseen scenes without pre-built maps [4, 5, 6] or costly scene-specific retraining [7, 8]. Zero-shot object navigation

gation (ZSON) formalizes this setting by requiring a robot to locate a specified object in a novel environment without task-specific fine-tuning [9, 10].

Despite strong results in simulation, current ZSON systems often degrade in cluttered real-world scenes with severe occlusions and latent hazards [11, 10, 12, 13]. We define a latent hazard as an initially unobserved, unannotated region that may cause collision, unsafe traversal, or task failure if traversed [14, 15, 16]. In these scenes, targets and risks may only become visible from revealing viewpoints, as illustrated by the occluded cat in Figure 1. Yet many existing methods act on a single inferred scene interpretation, making them prone to missing hidden targets or choosing unsafe paths through uncertain regions [17, 18].

To address this limitation, we draw inspiration from Schrödinger’s thought experiment on uncertainty and propose **Schrödinger’s Navigator**, a principled navigation framework that treats unobserved space as a superposition of plausible future worlds and reasons over them before committing to an action, as illustrated in Figure 1. Rather than completing the scene once, the agent reasons at inference time over multiple *trajectory-conditioned imagined 3D futures*. This enables the robot to evaluate how the world may appear along different candidate paths and plan not only from what is observed, but also from what could be revealed behind occlusions.

At the core of our framework is a trajectory-conditioned 3D world model [19], which differs from prior world-model navigation methods [20, 21, 22] in how imagined futures are generated and used. Given egocentric visual observations and sampled trajectories, the model predicts hypothetical 3D observations along each path, maintaining multiple plausible realizations of unobserved space. To make imagination efficient and decision-relevant, we introduce an *adaptive, occluder-aware trajectory sampling* strategy that uses local geometry and uncertainty to focus predictions on regions where occlusion is most likely to affect navigation, avoiding costly dense rollouts over the full scene. The imagined 3D predictions are spatially aligned and fused into a *Future-Aware Value Map* (FAVM). Unlike value maps built only from observed states or short-horizon predictions, FAVM aggregates evidence across candidate futures and assigns value to currently unobserved regions according to their expected utility. The policy can reduce uncertainty, expose likely target locations, and avoid hazardous regions without requiring dense global reconstruction or extra exploratory detours.

We evaluate Schrödinger’s Navigator in Habitat and on a physical Go2 quadruped across ObjectNav and occlusion-heavy scenarios. Results show that future-aware reasoning preserves competitive ObjectNav performance while improving behavior when targets or risks are initially hidden. By reasoning over imagined 3D futures at inference time, our method helps reveal hidden objects, avoid uncertain blind spots, and generalize to zero-shot navigation under real-world uncertainty.

In summary, our contributions are as follows:

- We identify a fundamental limitation of existing zero-shot object navigation methods under severe occlusion and latent hazards: their reliance on single-scene inference without explicit reasoning over uncertainty in unobserved space.
- We propose **Schrödinger’s Navigator**, a belief-aware navigation framework that reasons over multiple *trajectory-conditioned imagined 3D futures*. The framework combines a trajectory-conditioned 3D world model, adaptive occluder-aware trajectory sampling, and a *Future-Aware Value Map* (FAVM) for proactive uncertainty-aware planning.
- We demonstrate in both Habitat simulation and real-world Go2 deployment that explicit reasoning over imagined 3D futures improves robustness in occlusion-heavy object search, earlier hidden-target discovery, and risk-aware waypoint selection.

2 Related Work

Object Navigation (ObjectNav) [23] studies embodied agents that locate target objects in unseen environments. *Task-trained* approaches based on reinforcement or imitation learning [8, 1, 24, 25] require large-scale training and often generalize poorly to cluttered real-world scenes, making sim-

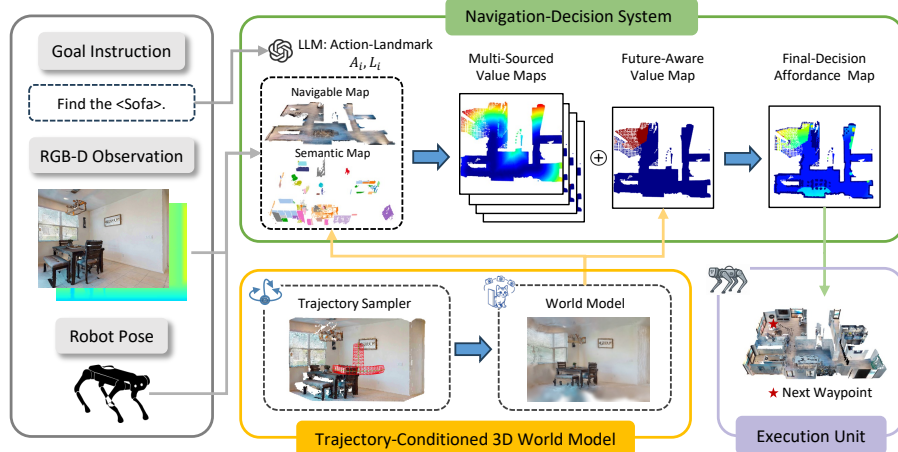


Figure 2: Overview of the Schrödinger’s Navigator. Given the navigation context, the system adaptively samples candidate trajectories around occluded and uncertain regions. A trajectory-conditioned 3D world model then imagines future 3DGS observations along each path to expose hidden structures, target cues, and risks. These imagined futures are fused with current observations to construct a Future-Aware Value Map, aggregating trajectory-level utility and risk for robust waypoint selection. Finally, the robot executes the selected waypoint via continuous control.

71 to-real deployment challenging. *Zero-shot* approaches instead leverage pretrained vision-language
 72 models [26, 9, 10] or large language models [27, 28, 29] for reasoning and planning without task-
 73 specific training [12, 10, 29, 30]. These methods improve semantics-driven exploration through mul-
 74 timodal target embeddings [9, 10], vision-language frontier maps [12], instruction-based prompting
 75 [31, 28], and adaptive fusion of semantic and geometric cues [32, 33, 34, 35, 36]. However, zero-
 76 shot ObjectNav remains brittle in realistic, occlusion-heavy environments and under unknown risks
 77 [10, 12], because value estimation is still dominated by visible frontiers and lacks explicit reasoning
 78 about hidden areas behind occluders, leading to myopic exploration.

79 **Imagination for Navigation** leverages generation models to simulate future observations and sup-
 80 port planning [37, 20, 38, 21]. Related world-model methods learn predictive dynamics and roll
 81 out trajectories in latent space [39, 40]. Recent navigation world models further predict egocentric
 82 visual streams. NWM [20] predicts future trajectories and views for path ranking, while Navi-
 83 gateDiff [38] uses diffusion-based visual prediction as a zero-shot navigation assistant. Language-
 84 conditioned variants, such as VISTA and VISTAv2 [41, 42, 22], generate landmark images or
 85 instruction-conditioned future views and use their alignment to guide action selection. Other works
 86 imagine unseen scenes or predict occupancy maps to complete unobserved spaces for exploration
 87 [43, 44, 45, 46, 47]. These works are mainly studied under visual prediction, or vision-and-language
 88 navigation protocols. Unlike prior works, Schrödinger’s Navigator imagines future observations
 89 along multiple paths with a trajectory-conditioned 3D world model and constructs a FAVM, enabling
 90 explicit reasoning about occlusions, risks, and blind-spot targets for robust zero-shot navigation.

91 3 Method

92 We introduce **Schrödinger’s Navigator**, a zero-shot object navigation framework that reduces
 93 occlusion-induced uncertainty by reasoning over multiple trajectory-conditioned imagined 3D fu-
 94 tures. At each decision step, the robot receives the current navigation context, including the goal
 95 instruction, RGB-D observation, robot pose, and accumulated map. Instead of committing to a
 96 single inferred scene, our method samples a compact set of occluder-aware candidate trajectories,
 97 imagines future 3D scenes along them, and aggregates the resulting cues into a future-aware affor-
 98 dance map for waypoint selection.

3.1 Trajectory-Conditioned 3D Future Imagination

Adaptive occluder-aware trajectory sampling. A single imagined path may miss targets or hazards hidden behind occlusions. To obtain complementary views with limited computation, we sample a small trajectory ensemble around the dominant occluder in the view. Specifically, we construct a local point cloud from the depth observation, remove ground points, and select the primary forward occluder according to projected occupancy, distance, and alignment with the robot’s heading. Its boundary provides a local geometric anchor for trajectory generation.

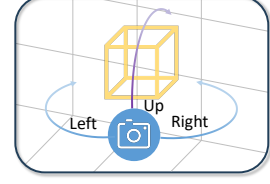


Figure 3: Occluder-aware camera trajectory sampling.

As shown in Figure 3, we instantiate three trajectory types $\mathcal{V} = \{L, R, U\}$, corresponding to left-bypass, right-bypass, and over-the-top views. For each type $v \in \mathcal{V}$, the sampler generates a camera trajectory $\tau_t^{(v)} = \mathcal{F}_{\text{samp}}(v; B_t, q_t, d_c, N_{\text{pose}})$, where B_t is the estimated occluder boundary, q_t is the current robot pose, d_c is a safe clearance distance, and N_{pose} is the number of sampled camera poses. This compact trajectory set is designed to cover both lateral and vertical reveal directions while avoiding dense rollouts over the full action space. Detailed boundary extraction, occluder selection, and trajectory parameterization are provided in Appendix B.

Trajectory-conditioned 3D world model. As shown in Figure 4, given each sampled trajectory $\tau_t^{(v)}$, we use a trajectory-conditioned 3D world model Φ_θ to generate a hypothetical future 3DGS:

$$\hat{\mathcal{G}}_t^{(v)} = \Phi_\theta \left(I_t, D_t, q_t, \tau_t^{(v)} \right),$$

where I_t and D_t denote the current RGB and depth observations. We adopt FlashWorld [19] as the backend because it can generate 3D-consistent scenes efficiently. Each generated scene represents a plausible realization of the currently unobserved region conditioned on one candidate path.

Since the generated 3DGS scene may not be metrically aligned with the robot’s environment, we align it to the current observation before fusion. We first render a depth map $\hat{D}_t^{(v)}$ from the generated scene at the current viewpoint and estimate a global scale factor using valid pixels:

$$s_t^{(v)} = \text{median}_{p \in \Omega_t^{(v)}} \frac{D_t(p)}{\hat{D}_t^{(v)}(p)},$$

where $\Omega_t^{(v)}$ is the set of pixels with valid observed and rendered depths. The generated scene is then scaled and transformed into the world frame as $\tilde{\mathcal{G}}_t^{(v)} = \mathcal{A}(\hat{\mathcal{G}}_t^{(v)}, s_t^{(v)}, q_t)$, where $\mathcal{A}(\cdot)$ denotes scale alignment and coordinate transformation. The aligned scenes are fused into a unified future scene $\mathcal{G}_t^{\text{img}} = \bigcup_{v \in \mathcal{V}} \tilde{\mathcal{G}}_t^{(v)}$. We further transfer semantic labels from 2D segmentation to visible Gaussian primitives using depth-consistent projection and multi-view voting. Implementation details of 3DGS alignment, semantic label transfer, and scene fusion are provided in Appendix C.

3.2 Future-Aware Navigation Decision

Future-Aware Value Map. The imagined scene $\mathcal{G}_t^{\text{img}}$ provides cues about free space, possible target locations, and risky regions beyond the current observation. We convert these cues into a *Future-Aware Value Map* (FAVM), denoted by m_t^{FA} , over navigable Gaussian primitives $g \in \mathcal{G}_{\text{nav}}$:

$$m_t^{\text{FA}}(g) = \alpha_{\text{sem}} S_t(g) + \alpha_{\text{exp}} E_t(g) - \alpha_{\text{risk}} R_t(g),$$

where $S_t(g)$ measures semantic relevance to the target, $E_t(g)$ measures the expected information gain from revealing newly imagined free space, and $R_t(g)$ penalizes proximity to predicted obstacles, hazards, or high-uncertainty regions. The weights α_{sem} , α_{exp} , and α_{risk} balance target seeking, exploration, and safety.

The three scores are computed from both observed and imagined Gaussian primitives within a local neighborhood of each candidate waypoint. Specifically, $S_t(g)$ is a distance-weighted target relevance score aggregated from target-labeled Gaussians, where imagined target cues are discounted

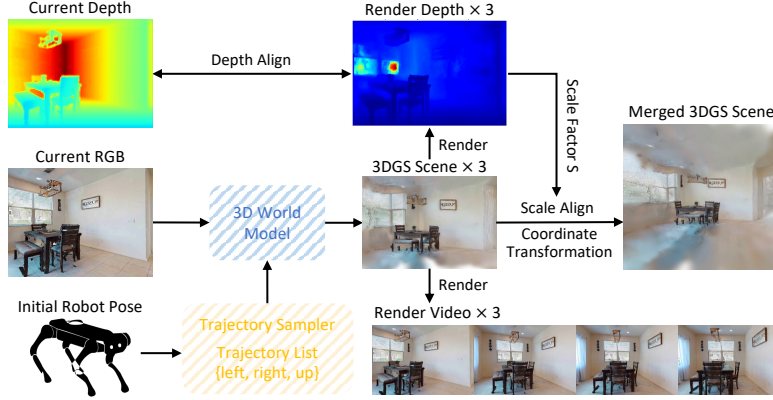


Figure 4: Overview of trajectory-conditioned 3D imagination: Given the current observation and robot pose, an occluder-aware sampler generates candidate trajectories that condition a 3D world model to predict future 3DGS scenes. These scenes are depth-aligned to recover a consistent scale, transformed into the world frame, and fused into a unified 3DGS scene for future-aware planning.

by a factor $\lambda_{\text{img}} < 1$ to avoid over-trusting generated content. $E_t(g)$ estimates the amount of previously unobserved but likely navigable space that can be revealed from g , computed from imagined free-space primitives not yet covered by the observed map. $R_t(g)$ is a soft speculative traversal penalty obtained by aggregating nearby imagined occupied or low-confidence primitives along the candidate path. All three scores are normalized to $[0, 1]$ before fusion; implementation details, including kernel radius, semantic voting, and normalization, are provided in Appendix D.

Fusion with observation-based cues. FAVM provides foresight, but navigation must remain grounded in current observations. We combine FAVM with an observation-based multi-sourced value map $m_t = m_t^{\text{act}} + m_t^{\text{sem}} + m_t^{\text{traj}} + m_t^{\text{heu}}$, which aggregates action guidance, semantic landmark cues, trajectory constraints, and heuristic navigation priors.

The final affordance map is computed as

$$m_t^{\text{aff}}(g) = \beta \bar{m}_t(g) + (1 - \beta) m_t^{\text{FA}}(g), \quad g \in \mathcal{G}_{\text{nav}},$$

where $\bar{m}_t$ is the normalized observation-based value map and $\beta \in [0, 1]$ controls the balance between grounded evidence and imagined future guidance. The next waypoint is selected as $p_t^* = \arg \max_{p \in \mathcal{P}_t} m_t^{\text{aff}}(p)$, where $\mathcal{P}_t$ denotes candidate navigable waypoint positions. Detailed definitions of FAVM scores and observation-based component maps are provided in Appendix D.

Execution. Following an instruction-driven hierarchical planner, the goal instruction is parsed into high-level action-landmark guidance, which contributes to the observation-based value map. The selected waypoint p_t^* is then passed to a local planner that generates high-frequency velocity commands for continuous robot control. To reduce latency, the world model is not invoked at every step; it is triggered only when a dominant forward occluder is detected, recent exploration or semantic progress is low, or a fixed refresh interval is reached.

4 Experiments

We evaluate Schrödinger’s Navigator in both simulation and real-world settings to assess its navigation performance, component effectiveness, and practical deployability. Our experiments first compare against strong ZSON baselines on HM3D, then ablate the key design choices behind future imagination and FAVM-based planning, and finally validate the system on a physical Go2 quadruped in indoor environments. More qualitative results and failure cases are shown in Appendix F.

Table 1: **Quantitative Comparison on Simulation Results.** Cell background colors indicate the method is the **best**, **second best**, or **third best** on this metric.

Method	Training Free	HM3D		
		SR↑	SPL↑	DTG↓
ZSON [9]	✗	0.255	0.126	–
PixNav [13]	✗	0.379	0.205	–
SPNet [48]	✗	0.312	0.101	–
SGM [35]	✗	0.602	0.308	–
ESC [28]	✓	0.392	0.223	–
VLFM [12]	✓	0.525	0.304	–
VoroNav [51]	✓	0.420	0.260	–
L3MVN [49]	✓	0.504	0.231	4.43
TriHelper [50]	✓	0.565	0.253	3.87
GAMap [33]	✓	0.531	0.260	–
InstructNav [31]	✓	0.510	0.187	2.89
ImagineNav [46]	✓	0.530	0.238	–
ImagineNav++ [47]	✓	0.625	0.328	–
BeliefMapNav [30]	✓	0.614	0.306	–
Ours	✓	0.652	0.311	2.23

4.1 Experimental Setup

Baselines and benchmark. We compare our approach against a broad set of representative baselines, including task-trained approaches (ZSON [9], PixNav [13], SPNet [48], SGM [35]), zero-shot semantic or map-based explorers (CoW [10], ESC [28], L3MVN [49], TriHelper [50], VoroNav [51], GAMap [33], BeliefMapNav [30]), imagination-based VLM navigators (ImagineNav [46], ImagineNav++ [47]), and LLM/VLM-based planners (VLFM [12], InstructNav [31]). We conduct large-scale simulation evaluation on HM3D [52] within Habitat [53]. Following prior work, we report Success Rate (SR), Success weighted by Path Length (SPL), and Distance to Goal (DTG). Detailed descriptions of baselines are provided in Appendix E.

Implementation. We use GPT-4o [54] for instruction parsing and navigation-direction judgment, FlashWorld [19] for trajectory-conditioned 3D imagination, and GLEE [55] for semantic segmentation. FlashWorld is triggered periodically or under significant occlusion, rather than at every step. Real-world tests use a Unitree Go2 with an RGB-D camera in Office, Classroom, and Common Room scenes, covering static-object search and occlusion-heavy unknown-risk cases. Hyperparameters, runtime, compute, and deployment details are provided in Appendices A and G.

4.2 Simulation Results

4.2.1 Quantitative Results

Table 1 reports the quantitative results on HM3D. Our method achieves the best SR among all compared methods and the lowest DTG among methods that report this metric, indicating that future-aware planning improves both target discovery and final goal proximity. This suggests that anticipating future observations helps the agent select more promising exploration directions, rather than relying only on reactive perception or static belief estimation. Although our method ranks second in SPL, it remains competitive with both task-trained and training-free methods, reflecting a reasonable trade-off between success and path efficiency: additional exploratory actions may slightly increase path length, but they lead to more reliable object localization. Overall, the high SR, low DTG, and competitive SPL demonstrate that our method improves navigation reliability while maintaining strong path efficiency in zero-shot object navigation.

Table 2: Ablation and runtime analysis of Schrödinger’s Navigator on HM3D. We report success rate (SR), average number of imagination triggers per episode, per-trigger imagination latency, episode-level waiting ratio, and waypoint change rate.

Variant	SR $\uparrow$	Trig. / Ep. $\downarrow$	Lat. / Trig. (s) $\downarrow$	Wait Ratio (%) $\downarrow$	Waypoint Change (%)
No Imagination	0.553	0.00	–	0.0	–
Single Traj. ($N = 1$)	0.582	4.31	3.0	7.2	28.4
Dense Sampling ($N = 5$)	0.636	4.18	15.0	30.8	51.6
Imagination-heavy ($\beta = 0.2$)	0.450	4.36	9.0	22.9	67.2
Observation-heavy ($\beta = 0.8$)	0.604	4.12	9.0	18.6	30.5
FAVM w/o Exploration $E(g)$	0.615	4.20	9.0	19.8	40.8
FAVM w/o Semantic $S(g)$	0.599	4.27	9.0	20.5	38.4
FAVM w/o Risk $R(g)$	0.617	4.19	9.0	19.3	42.1
Ours ($N = 3, \beta = 0.5$)	0.652	4.24	9.0	20.1	46.8

4.2.2 Ablation Analysis

To assess the key design choices of Schrödinger’s Navigator, we ablate the trajectory ensemble size N , the fusion coefficient β , and the FAVM scoring components. All variants change one factor from the full model, except for the no-imagination baseline.

The results in Table 2 show that future imagination is effective only when properly sampled and grounded. Our full model improves SR from 0.553 to 0.652 over the no-imagination baseline. However, more trajectories do not necessarily yield better performance: $N = 1$ provides limited coverage, while $N = 5$ incurs higher latency and still underperforms our $N = 3$ design, demonstrating the coverage–cost advantage of tri-trajectory sampling.

The fusion and component ablations further show that imagined guidance must be calibrated by real observations. The imagination-heavy setting ($\beta = 0.2$) drops to 0.450, suggesting that over-trusting generated content can mislead planning, whereas the observation-heavy setting ($\beta = 0.8$) is more stable but less effective than the balanced setting. Removing any FAVM component reduces SR, with the largest drop caused by removing the semantic score $S(g)$.

The runtime results indicate that 3D future imagination introduces sparse rather than per-step overhead. Our full model triggers imagination 4.24 times per episode, with 9.0 seconds per trigger and a 20.1% waiting-time ratio. Dense sampling increases the waiting-time ratio to 30.8% without improving SR, showing that additional futures introduce extra cost without proportional benefit. The 46.8% waypoint change rate further indicates that imagined futures actively influence waypoint selection beyond the observation-only map.

4.3 Real-World Deployment

We evaluate adaptability under two task settings: (1) **Static Object Search**, where the agent explores standard scenes to locate objects; and (2) **Unknown-Risk Scenarios**, where severe static occlusions are not visible from afar, testing the agent’s ability to anticipate uncertainty and avoid blind spots.

4.3.1 Performance Analysis

As shown in Table 3, we report success counts over 30 total trials.

In standard static object search, Schrödinger’s Navigator performs comparably to InstructNav, achieving a slightly higher success rate (23/30 vs. 22/30), indicating that future-aware reasoning does not compromise regular object-search performance. In scenarios with unknown risks, our method shows a clearer advantage, succeeding in 17/30 trials compared with InstructNav’s 11/30.

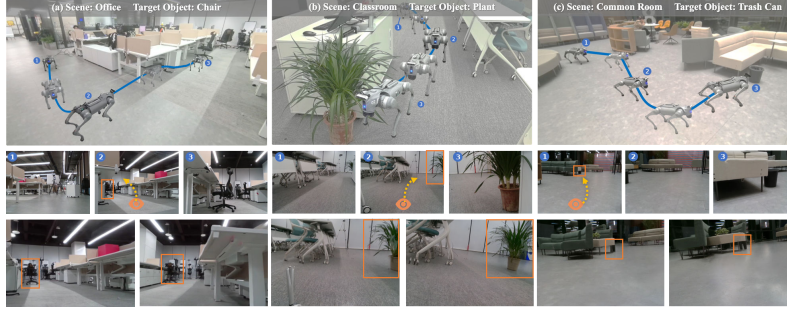


Figure 5: Real-world object-navigation demonstrations of Schrödinger’s Navigator across three indoor scenarios: Office-Chair, Classroom-Plant, and Common Room-Trash Can. Each example shows the executed trajectory, egocentric observations with predicted target directions and target boxes, and world-model imagined views that reveal the target appearance.

Table 3: Real-world quantitative comparison against the InstructNav baseline. We report success counts (Success / Total Trials) across three indoor environments.

Scene	Office	Classroom	Common Room	All
Search for static objects				
InstructNav	7/10	7/10	8/10	22/30
Ours	8/10	8/10	7/10	23/30
Scenarios with Unknown Risks				
InstructNav	4/10	3/10	4/10	11/30
Ours	5/10	6/10	6/10	17/30

225 This suggests that imagined 3D futures help the robot better handle occlusions, ambiguous observa-
 226 tions, and potential blind spots.

227 Overall, these real-world results show that our method maintains strong static-search performance
 228 while improving robustness in risk-aware navigation. Qualitative examples are provided in [Figure 5](#).

229 5 Conclusion

230 Schrödinger’s Navigator addresses ZSON limitations in occluded environments by leveraging a 3D
 231 trajectory-conditioned world model to imagine and fuse plausible 3D futures into a unified scene.
 232 This training-free framework constructs a single affordance map encoding semantics, exploration,
 233 and risk. Real-world experiments on a Unitree Go2 demonstrate its robustness, matching baselines
 234 in static tasks while significantly outperforming them in complex scenarios.

235 6 Limitations

236 Schrödinger’s Navigator uses imagined 3D futures as plausible hypotheses rather than exact re-
 237 constructions of hidden space. While FAVM discounts generated cues and grounds decisions in
 238 current observations, inaccurate generation, semantic transfer errors, or imperfect depth alignment
 239 may still bias waypoint selection when imagined evidence is over-weighted. Future work could im-
 240 prove reliability through uncertainty-calibrated imagination, stronger multi-view consistency, and
 241 active verification of uncertain regions. In addition, the current tri-trajectory sampler focuses on
 242 the dominant forward occluder for efficiency, but more complex layouts with multiple occluders or
 243 non-frontal blind spots may require adaptive trajectory budgeting. Broader evaluation across more
 244 diverse environments, dynamic obstacles, sensors, and robot platforms, as well as integration with
 245 certified local planners, would further strengthen real-world deployment.

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